Project Initialization and Planning Phase

Date	28-07-2026
Team ID	SWTID-2026-3845
Project Title	SMART LENDER
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To develop a machine learning-powered web application that predicts the creditworthiness of loan applicants, enabling banks and financial institutions to make faster, data-driven loan approval decisions.
Scope	The project covers data preprocessing, training and comparison of multiple classification algorithms, selection of the best-performing model, and integration into a Flask-based web application for real-time predictions, designed for deployment on Render
Problem Statement	
Description	Traditional loan approval processes rely heavily on manual evaluation of applicant details such as income, credit history, and employment status
Impact	Manual and inconsistent credit evaluation increases the risk of loan defaults, delays approval turnaround time, and contributes to a rise in non-performing assets for banks and financial institutions.
Proposed Solution	
Approach	Build and compare multiple machine learning classification models on historical applicant data, select the most accurate model (XGBoost), and deploy it within a Flask web application.
Key Features	Real-time prediction through a simple web interface

	High-risk applicant detection for irregular income or missing credit history Support for fast-track approval of low-risk applicants

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv